

PHYS 225

Week 6, Lecture 6:
Symmetries and Group Theory



UNIVERSITY OF
ILLINOIS
URBANA-CHAMPAIGN

Reminders

- Homework 5 due at 11:59 pm this Thursday, 2/26.
- In person Office Hours **Wednesday/Thursday**, check the website, OH page, for additional online options.
- **Midterm is on 3/10 at 4pm, rooms Loomis 141 and tba**
 - Same general format/scope as homework, but shorter.
 - You are allowed a double-sided 8.5 x 11 page of your own formulas/notes, plus calculators.
 - A practice exam will be posted **this week**.
 - If you need DRES accommodations, please set them up the week before! (The night before is too late to accommodate).
- Don't forget to turn in your **1-minute papers** by 6 pm today.

Week 6, Lecture 6

Symmetries and Group Theory

Objectives:

- Raise and lower indices with contractions
- Learn the concepts of transpose, inverse, and determinant
- Define an orthogonal matrix as one which preserves the dot product
- Define a mathematical group
- Show that Lorentz boosts along a single axis form a group

One Minute Paper: how would you define the concept of “symmetry”?

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Symmetries and Group Theory

A note before we begin

The material we will cover today includes some crucial concepts in physics and mathematics, but it will **not** be covered on the midterm, so you have some time to study it.

(It will show up in HW and final exam, though!)

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Types of vectors

Let's distinguish between two types of vectors.

Column vector: $v^i = \begin{pmatrix} v^1 \\ v^2 \end{pmatrix}$ index goes **upstairs**

Row vector: $w_i = (w_1 \quad w_2)$ index goes **downstairs**

Recall: M^i_j  labels row  labels column $= \begin{pmatrix} M^1_1 & M^1_2 \dots \\ M^2_1 & M^2_2 \dots \\ \vdots & \ddots \end{pmatrix}$

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Types of vectors

Let's distinguish between two types of vectors.

Column vector: $v^i = \begin{pmatrix} v^1 \\ v^2 \end{pmatrix}$ index goes upstairs

Row vector: $w_i = (w_1 \quad w_2)$ index goes downstairs

Dot product is really a product of a row vector and a column vector
(the arrow notation hides this, which is why indices are better)

$$\vec{w} \cdot \vec{v} = (w_1 \quad w_2) \cdot \begin{pmatrix} v^1 \\ v^2 \end{pmatrix} = w_i v^i = w_1 v^1 + w_2 v^2$$

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Contractions

$$(MN)^i_k = M^i_j N^j_k$$

As we saw in Weeks 3 and 5, matrix or vector multiplication is a bunch of repeated dot products. Any such dot product has a summation over a pair of indices and is called a **contraction**:

$$M = \begin{pmatrix} M^1_1 & M^1_2 \\ M^2_1 & M^2_2 \end{pmatrix}$$

$$N = \begin{pmatrix} N^1_1 & N^1_2 \\ N^2_1 & N^2_2 \end{pmatrix}$$

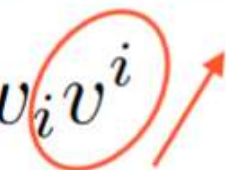
$$MN = \begin{pmatrix} M^1_1 N^1_1 + M^1_2 N^2_1 & M^1_1 N^1_2 + M^1_2 N^2_2 \\ M^2_1 N^1_1 + M^2_2 N^2_1 & M^2_1 N^1_2 + M^2_2 N^2_2 \end{pmatrix}$$

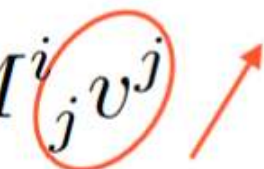
$$\begin{pmatrix} M^1_1 N^1_2 + M^1_2 N^2_2 \\ M^2_1 N^1_2 + M^2_2 N^2_2 \end{pmatrix}$$

Week 6, Lecture 6

Contractions

As we saw in Weeks 3 and 5, matrix or vector multiplication is a bunch of repeated dot products. Any such dot product has a summation over a pair of indices and is called a **contraction**:

$$\vec{w} \cdot \vec{v} \equiv w_i v^i$$


$$M\vec{v} \equiv M^i_j v^j$$


Remember, all contractions are “northeast” ↗

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Index gymnastics

In Lecture 3 you learned a fancy way to write dot products:

$$\vec{w} \cdot \vec{v} = \delta_{ij} w^i v^j = w^i (\delta_{ij} v^j)$$

Once we write indices, we can reorder the terms however we want!
The “northeast” convention preserves the order in matrix notation

Let's **define** the object in parentheses to be a row vector:

$$v_i \equiv \delta_{ij} v^j$$

Jargon: we **lowered an index** on v^j by **contracting** with δ_{ij}

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Index gymnastics

We can also lower an index on w^i :

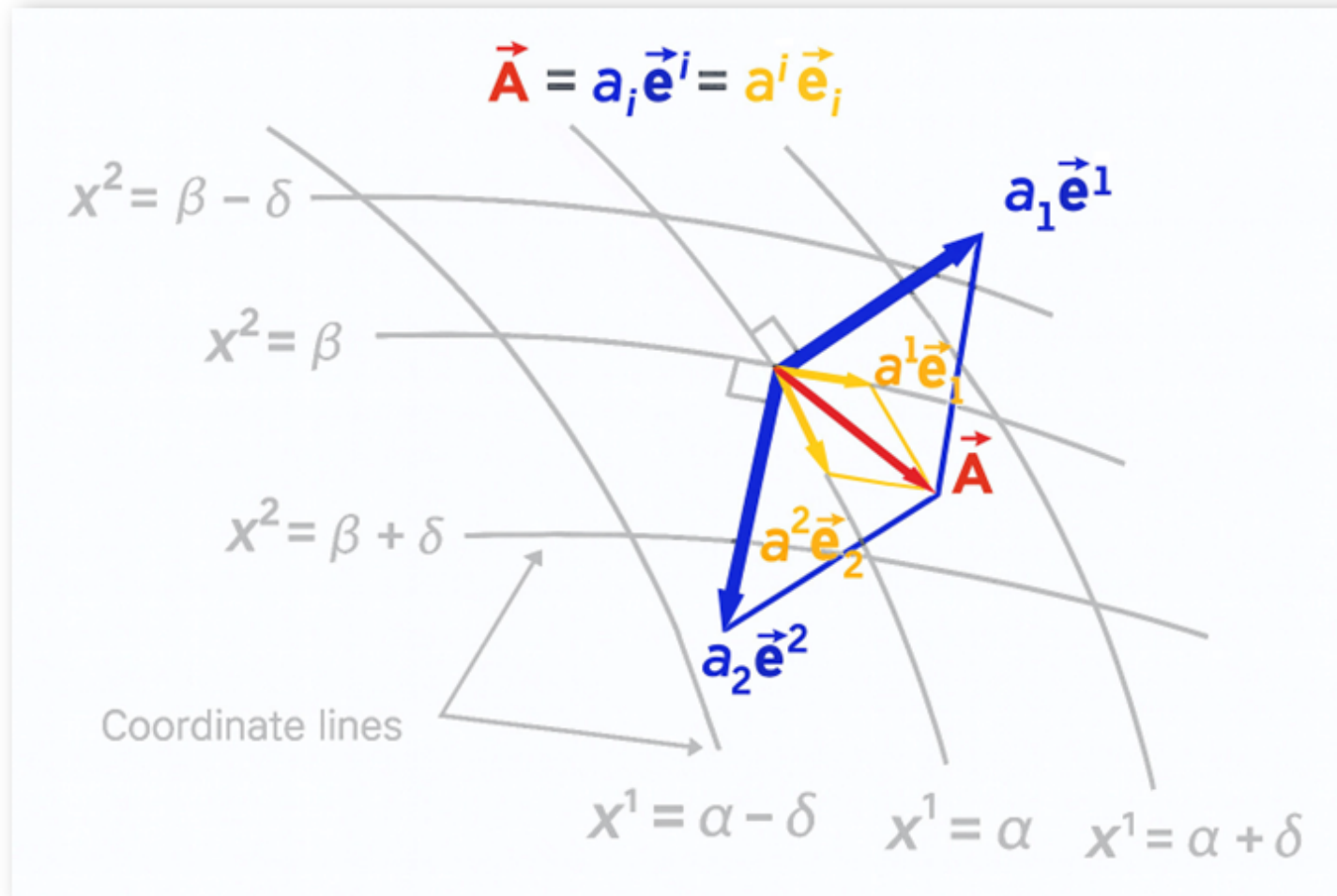
$$\vec{w} \cdot \vec{v} = \delta_{ij} w^i v^j = (\delta_{ij} w^i) v^j = w_j v^j$$

The dot product is **commutative** ($\vec{w} \cdot \vec{v} = \vec{v} \cdot \vec{w}$)
because the Kronecker delta is **symmetric**: $\delta_{ij} = \delta_{ji}$

Finally, let's define a version of the Kronecker delta with indices upstairs: $\vec{w} \cdot \vec{v} = \delta^{ij} w_i v_j$

This object lets us **raise an index**: $v^j \equiv v_i \delta^{ij}$

When indices matter – non-orthogonal basis $\{\vec{e}_i\}$
and its reciprocal basis $\{\vec{e}^j\}$ (Euclidean 2D plane example)



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Transposes of vectors

The operation “exchange rows and columns” is called the **transpose**

$$\vec{v} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$$

$$\vec{v}^T = (1 \quad 2 \quad 3)$$

applying it twice brings us back: $(\vec{v}^T)^T = \vec{v}$

Yet another way to think of the dot product:

$$\vec{w} \cdot \vec{v} \equiv \vec{w}^T \vec{v}$$

raising or lowering index
= taking transpose

$$\vec{w} \cdot \vec{v} \equiv w_i v^i$$

NOTE!! Transpose always goes first:
you can't dot a column into a row!



index notation: contraction is
always northeast

Week 6, Lecture 6

Transposes of matrices

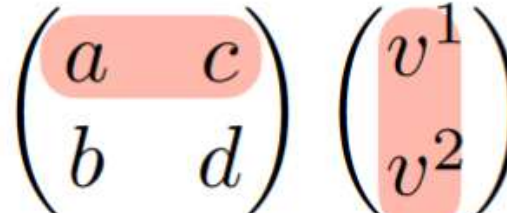
Can we multiply a row vector by a matrix?

Sure, but row vector must come first:

$$\vec{v}^T = (v_1 \quad v_2) \quad M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$


$$\vec{v}^T M = (av_1 + cv_2 \quad bv_1 + dv_2)$$

This is the same result as if we had multiplied a column vector by the **transpose** of the matrix in the usual way:

$$M^T \vec{v} = \begin{pmatrix} a & c \\ b & d \end{pmatrix} \begin{pmatrix} v^1 \\ v^2 \end{pmatrix} = \begin{pmatrix} av^1 + cv^2 \\ bv^1 + dv^2 \end{pmatrix}$$


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Transposes and indices

Taking transposes reverses order:

$$(\vec{v}^T M)^T = M^T \vec{v}$$

Also true for matrices (see Shankar 8.4):

$$(MN)^T = N^T M^T$$

Now with index notation:

$$(\vec{v}^T M)_j = v_i M^i_j = M^i_j v_i$$

row vector gets
a lower index

wrong contraction for Mv ??

Transpose (swapping upper and lower indices) fixes it:

$$(\vec{v}^T M)^T = M^T \vec{v} = M^j_i v^i$$

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Orthogonal matrices

Let's try multiplying two vectors by the **same** matrix M :

$$\vec{v}' = M\vec{v} \qquad \vec{w}' = M\vec{w}$$

Does the dot product stay the same?

$$\begin{aligned} \vec{v}' \cdot \vec{w}' &= (\vec{v}')^T \vec{w}' \\ &= (M\vec{v})^T (M\vec{w}) \\ &= \vec{v}^T (M^T M) \vec{w} \stackrel{?}{=} \vec{v}^T \vec{w} \end{aligned}$$

For arbitrary $\vec{v}$ and $\vec{w}$, equal if and only if $M^T M$ is the “do-nothing” identity matrix

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Orthogonal matrices

$$M^T M = I$$

Matrices satisfying this property are called **orthogonal**.

Orthogonal matrices preserve the dot product.

If performing an operation (multiplication by an orthogonal matrix)
leaves something (dot product) unchanged,
we call that operation a **symmetry**

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Matrix inverses

For some matrices M , there exists a matrix which can be multiplied by M to get the identity. If such a matrix exists, it is called the **inverse**, and given the name M^{-1}

$$M^{-1}M = I$$

$$(M^{-1})^i_j M^j_k = \delta^i_k$$

Just like transposes, applying inverse twice gets us back:

$$(M^{-1})^{-1} = M \implies MM^{-1} = I$$

[blackboard]

To show $(M^{-1})^{-1} = M$, let $(M^{-1})^{-1}$ be some matrix G . Then by the definition of inverses,

□

$GM^{-1} = I$. Multiply both sides on the right by M :

$$GM^{-1}M = IM$$

⏟ ⏟
I M

$$\Rightarrow G = M, \text{ so } GM^{-1} = I \Rightarrow MM^{-1} = I.$$

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Matrix inverses

Finding the inverse of a generic matrix is not easy!
Orthogonal matrices are special, though:

$$M^T M = I$$

$$M^{-1} M = I$$

$$M^T = M^{-1}$$

To invert an orthogonal matrix, just swap rows and columns!
(NOT true for a general matrix)

note we can now also write for an orthogonal matrix, $MM^T = I$

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Determinants

How do we know if a matrix has an inverse? For a square matrix, necessary and sufficient condition is that a number called the **determinant** is nonzero.

$$M = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \qquad \det M \equiv \begin{vmatrix} a & b \\ c & d \end{vmatrix} \equiv ad - bc$$

The formula for larger matrices is more complicated and we won't need it.

Most important properties of determinants (MATH 225/257):

$$\det(M^T) = \det M \quad \det I = 1 \quad \det(MN) = (\det M)(\det N)$$

$$M^T M = I \implies (\det M)^2 = 1 \implies \det M = \pm 1$$

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Groups

A collection of objects is called a **group** if it has an associative multiplication rule and satisfies the following 3 properties:

1. Identity there is a special element I such that $gI = Ig = g$
for all g in the collection
2. Inverse for every element g , there is another element g^{-1} such that
$$g^{-1}g = gg^{-1} = I$$
3. Closure if g and h are in the collection, then gh is in the collection

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Lorentz boosts form a group!

Let's show that Lorentz transformations along the x axis form a group.

1. Identity $I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ is a Lorentz transformation with $\beta = 0$

2. Inverse $\Lambda = \begin{pmatrix} \gamma & \gamma\beta \\ \gamma\beta & \gamma \end{pmatrix}, \Lambda^{-1} = \begin{pmatrix} \gamma & -\gamma\beta \\ -\gamma\beta & \gamma \end{pmatrix}$
(Discussion 5)

3. Closure Velocity addition! We did this last week.

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Symmetries and Group Theory

Summary

- Switch between row and column vectors by transpose, or contraction with Kronecker delta (raising and lowering indices)
- Dot products are really between row and column vectors
- Orthogonal matrices satisfy $M^T M = M M^T = I$, and preserve the dot product between two vectors
- Groups of matrices satisfy identity, inverse, closure
- Lorentz transformations form a group

One Minute Paper: does $M^i_j v_i$ represent $M \vec{v}$ or $\vec{v}^T M$?